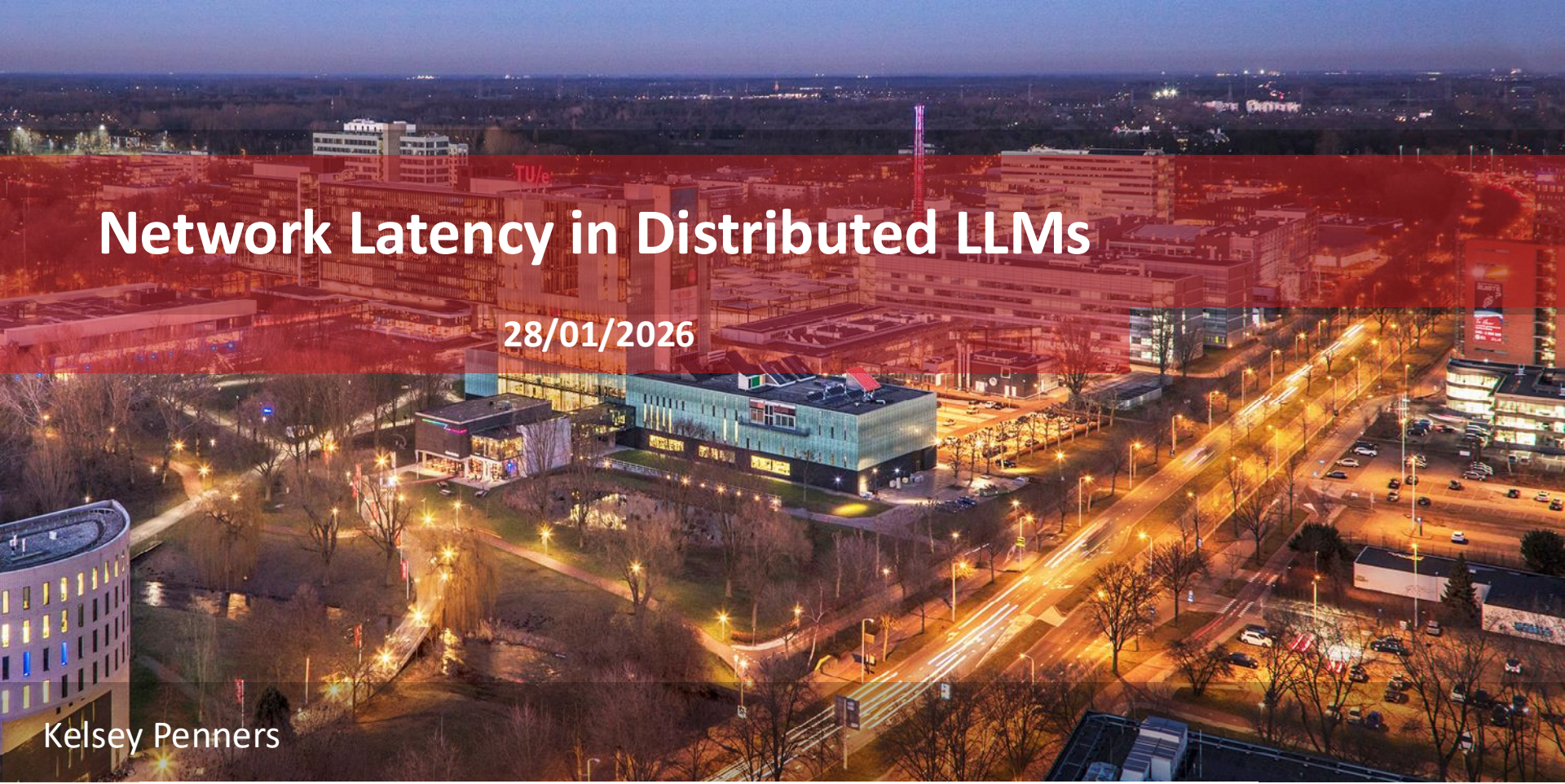


Network Latency in Distributed LLMs

28/01/2026

Kelsey Penners

TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

About Me!

Kelsey Penners

Master's Student | Artificial Intelligence & Engineering Systems | TU/e

Background:

- BSc in Computer Science at *University of California, Santa Cruz*
- Strong foundation in ML, software development, & data-driven systems
- Experience in building end-to-end ML pipelines and performance optimization

Current Project:

- 32-week master's thesis, supervised by Prof. Alex Alvarado
- Focus: Network latency impact on distributed LLM inference

Project Overview

Research Focus:
How does communication latency affect distributed LLM inference performance?

Scope:

- Study inference performance in single GPU, multi-GPU, and multi-node settings
- Systematically vary communication latency and network conditions
- Measure standard metrics (e.g., TTFT, throughput, inter-token latency)
- Compare performance as communication overhead increases
- Variables to explore: network latency ranges, parallelization strategies, model architectures

Timeline: Week 2 of 32 (currently in literature review and planning phase)